Tübingen AI Center

Image Processing
Exercise 1

WS 25 / 26

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Machine Learning Engineering and Technology Transfer

Due in two weeks 13.11.25

Exercise 1

- Project 1: Color2Gray
- Project 2: Maria-von-Linden
- Project 3: Creating Mosaics

Project 1 – Color2Gray

- RGB Color images have three channels, grey images only one
- Write a small script that uses pytorch Conv2d to convert a 3-channel image to 1-channel greyscale
- Ask for BT.601 weights (0.299, 0.587, 0.114)
- A sample call could be: `python convert.py input.png output.png`

Project 2 – Maria von Linden

- Check the floor of the building Maria-von-Linden-Strasse 1
- How many tiles are there?
- How would you use a linear filter and pytorch Conv2d to recreate the mosaic?
- Write a function that re-recreates the mosaic
- Image is attached

Project 3 – Photo Mosaics

Example:

https://www.flickr.com/photos/genista/6898950/in/set-172701/

Project 3 – Photo Mosaics

- Write a function / webpage that will accept an image and then create a mosaic version thereof
- What does that have to do with filtering/convolutions?
- What are possible ways to select the images? What do we need to compute?
- On ILIAS
- Bonus:
  - Allow to specify grid size (#columns, #rows)
  - Allow to use your own photo set
  - Have one additional method of selecting images